3.5.3 Transient Behavior:

Forward-to-reverse bias

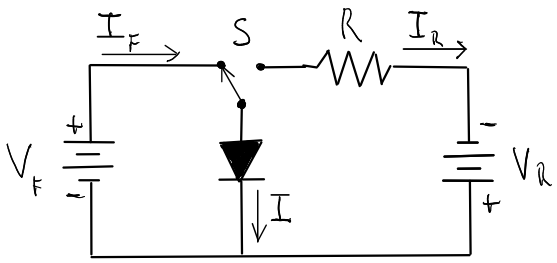
⇓

I_F transient time is short

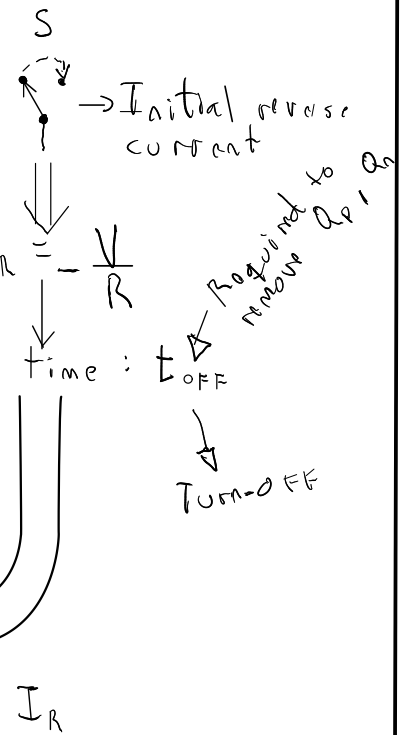
⇓

Switching apps.

Basic switching circuit:



At $t=0_S$:



For p^+-n junction, under forward condition:

⇓

$Q_n \simeq 0$

Stored minority carriers in the n-region (neutral):

$$Q_p = \left(\frac{L_p^2}{D_p}\right)\frac{q D_p}{L_p} p_{no}\left(e^{qV/kT}-1\right)$$

$$Q_p \simeq \tau_p J_{pdiff}(x=x_n) = \tau_p J_F$$

$J_{ndiff} \simeq 0$

During turn-OFF period: $I_{R,ave}$:

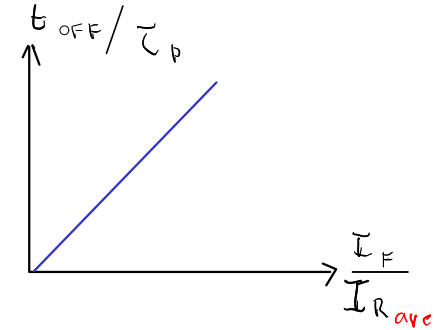
↳ average

$$I_{R,ave} = \frac{Q_p A}{t_{OFF}}$$

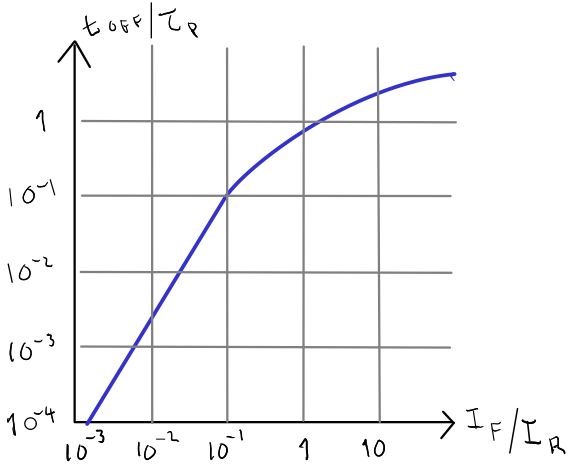
$$t_{OFF} \simeq \frac{Q_p A}{I_{R,ave}} \simeq \frac{(\tau_p J_F) A}{I_{R\,ave}}$$

$$t_{OFF} \simeq \tau_p \left(\frac{I_F}{I_{R\,ave}}\right)$$

$$\frac{t_{OFF}}{\tau_p} \simeq \frac{I_F}{I_{R\,ave}}$$



Minority-carrier diffusion → Time depend → More precise t_{OFF} calculation



Recombination centers at E_t near to E_i ($E_t \approx E_i$)

↓

$U_{máx}$

↓

τ_P

↓

t_{OFF} → Fast-switching devices

Under forward bias recombination rate in n region:

$$U = \frac{\sigma_0 v_{th} N_t n_i^2 \left(e^{qV/kT} - 1\right)}{n_n + p_n + 2 n_i \cosh\left(\frac{E_t - E_i}{kT}\right)}$$

$U_{máx}$ at $E_t \approx E_i$

3.6 JUNCTION BREAKDOWN

Large reverse voltage → It could reach I_{max}

↓ Breakdown process is not inherently destructive

↓ Excessive junction heating

↓ but

Breakdown mechanisms: ① Tunneling effect

② Avalanche multiplication

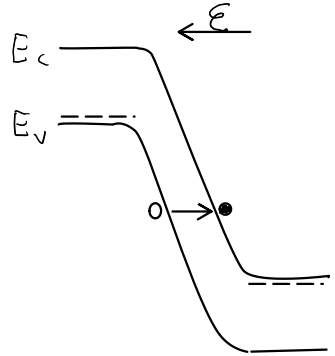
Also: Avalanche breakdown limits V_{DS} of MOSFET

I_F breakdown voltage $\approx \frac{4E_g}{q}$ ← Tunneling effect

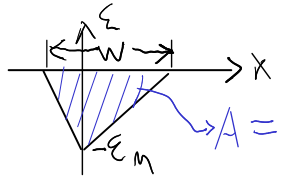
I_F breakdown voltage $\approx \frac{6E_g}{q}$ ← Avalanche multiplication

If breakdown voltage $\approx \frac{4E_g}{q}$ to $\frac{6E_g}{q}$ ← Mixture: Avalanche mult. Tunneling eff.

3.6.1 Tunneling Effect



For abrupt junction at V=0:



$$A = V_{bi} = \frac{\varepsilon_m W}{2}$$

$$\varepsilon_m = \frac{2V_{bi}}{W}$$

$$\varepsilon_m = 2V_{bi}\sqrt{\frac{q}{2\varepsilon_s}\left(\frac{N_A N_D}{N_A + N_D}\right)}$$

* For Si and GaAs:

$$\varepsilon > 10^6 \frac{V}{cm} = \frac{5V}{50nm} = \frac{9V}{10nm}$$

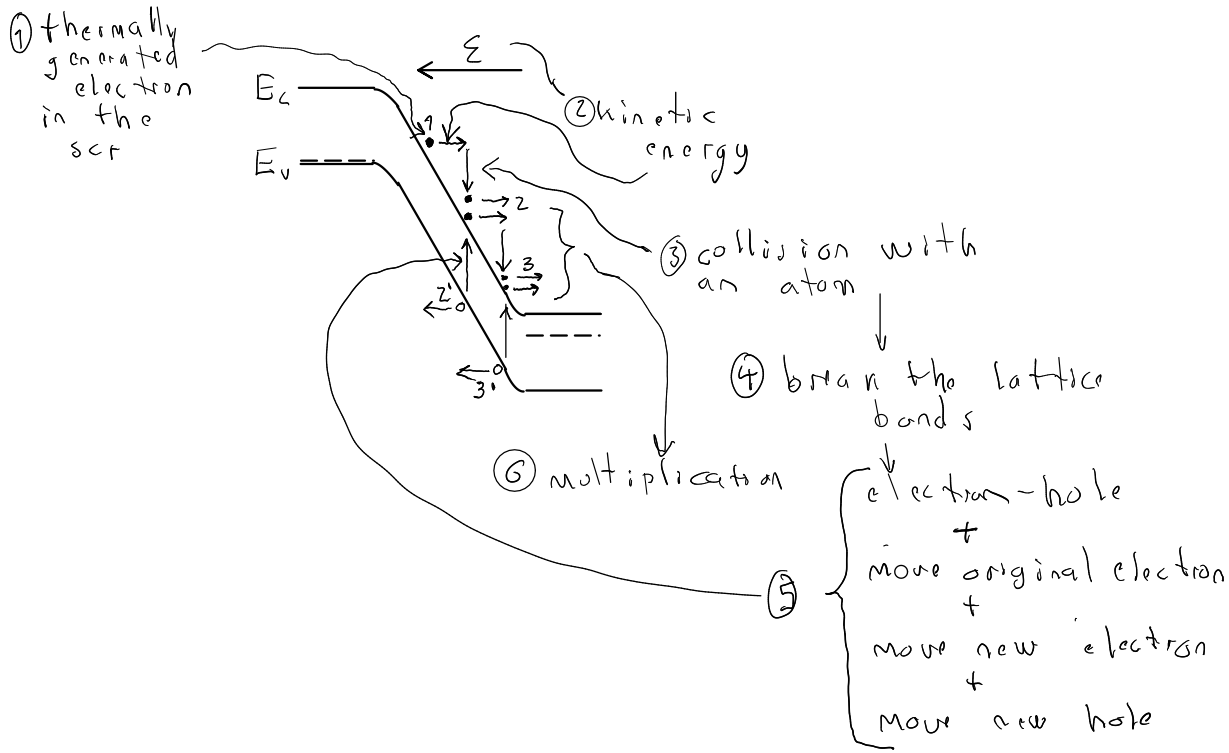
↓

Tunneling effect ←

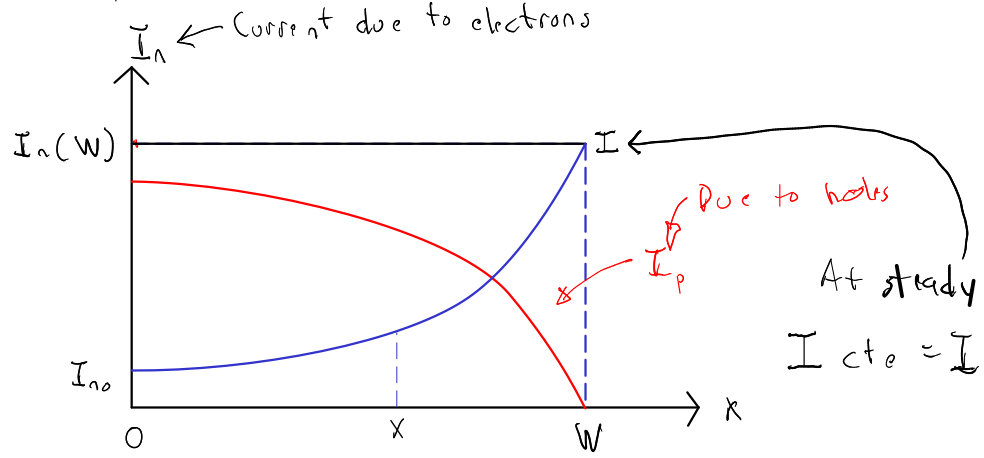
$(N_A \& N_D) > 5 * 10^{17} cm^{-3}$ → achive high ε

3.6.2 Avalanche Multiplication

p^+-n junction (one-sided abrupt): $N_D < 10^{17} cm^{-3}$ under reverse bias:



* For $p^+ - n$:



At steady state :

$$I_{cte} = I_n + I_p = I_n(W)$$

Multiplication factor :

$$M_n = \frac{I_n(W)}{I_{no}}$$

$$I_n(W) = M_n I_{no}$$

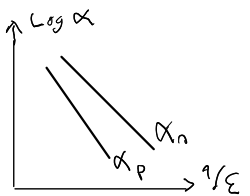
Ionization rate (Chap 2.8)

$$\alpha_n = \frac{\#\ \text{electron-hole pair generated}}{\text{Distance traveled by an electron}}$$

$$\alpha_p = \frac{\#\ \text{electron-hole pair generated}}{\text{Distance traveled by an hole}}$$

$$[\alpha] = cm^{-1}$$

$$\alpha(\mathcal{E}) \propto \mathcal{E}$$



$$I_{\text{incremental electron current at } x} = \frac{\#\ e\text{-}h \text{ generated}}{\text{time}}$$

$$d\left(\frac{I_n}{q}\right) = \left(\frac{I_n}{q}\right)(\alpha_n dx) + \left(\frac{I_p}{q}\right)(\alpha_p dx)$$

(α_n: ionized electrons; α_p: ionized holes)

$$dI_n - I_n \alpha_n dx = I_p \alpha_p dx$$

$$\frac{dI_n}{dx} - I_n \alpha_n = (I - I_n)\alpha_p$$

$$\frac{dI_n}{dx} + (\alpha_p - \alpha_n) I_n = \alpha_p I$$

If : $\alpha_n = \alpha_p = \alpha$

$$\frac{dI_n}{dx} = \alpha I$$ ← I cte at steady state

$$\frac{dI_n}{I} = \alpha dx$$

$$\frac{I_n(W) - I_{no}}{I} = \int_0^W \alpha dx$$

$$1 - \frac{1}{M_n} = \int_0^W \alpha \, dx$$

Avalanche breakdown $\leftarrow$ $M_n \to \infty$

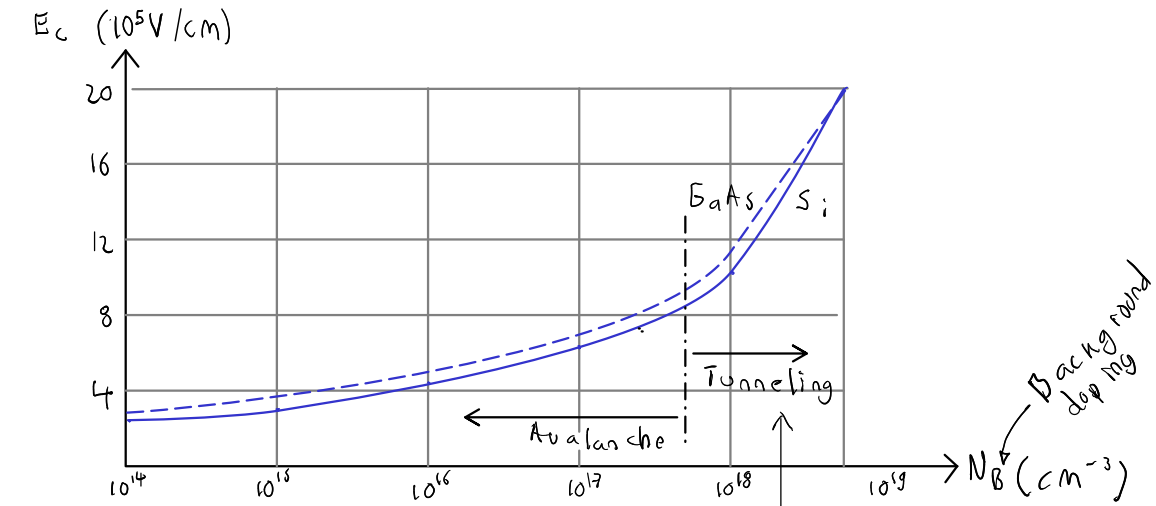
So:

$$\boxed{1 = \int_0^W \alpha \, dx}$$ $\leftarrow$ Breakdown condition

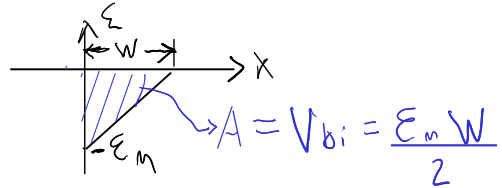
Breakdown condition & $\alpha(\varepsilon)$ $\longrightarrow$ Calculate the critical field (ε_{min} at breakdown)

$1 = \int_0^W \alpha(\varepsilon) dx \longrightarrow \varepsilon_{min} \approx \varepsilon_c \longrightarrow$ Avalanche process

* For one-sided abrupt junctions:



For p^+-n junction at $V=0$:



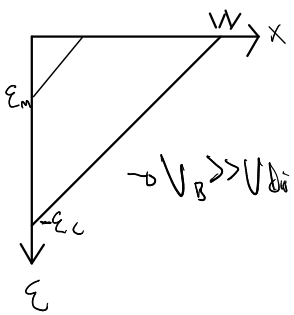
$$A = V_{bi} = \frac{\varepsilon_m W}{2}$$

$$\varepsilon_m = \frac{2 V_{bi}}{W}$$

$$\varepsilon_m = 2 V_{bi} \sqrt{\frac{q N_D}{2 \varepsilon_s}}$$

ε_m

only high doping concentrations ($N_B > 5 * 10^{17}$ cm^{-3})



For p⁺-n at breakdown voltage:

$$V_B + V_{bi} = \frac{\mathcal{E}_c^2 \varepsilon_s}{2q N_B}$$

→ $V_B >> V_{bi}$

$$V_B \overset{?}{\approx} \frac{\mathcal{E}_c^2 \varepsilon_s}{2q N_B}$$

For p⁺n at V=0:

$$\mathcal{E}_m = \frac{q N_B W}{\varepsilon_s}$$

$$W = \frac{\varepsilon_s \mathcal{E}_m}{q N_B}$$

$$V_{bi} = \frac{\mathcal{E}_m W}{2}$$

$$V_{bi} = \frac{\mathcal{E}_m^2 \varepsilon_s}{2q N_B}$$

*For linearly graded junction (Chap 3.2.2):

$$\int_0^{V_{bi}} V\, dV = -\int_{-W/2}^{W/2} \mathcal{E}\, dx \quad \Longleftrightarrow \quad \mathcal{E} = -\nabla V$$

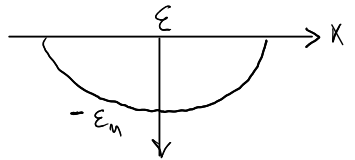
$$V_{bi} = \frac{qa}{2\varepsilon_s}\left[\frac{W^2}{4}x\Big|_{-W/2}^{W/2} - \frac{x^3}{3}\Big|_{-W/2}^{W/2}\right]$$

$$V_{bi} = \frac{qa}{2\varepsilon_s}\left[\frac{W^2}{4}W - \frac{1}{3}\left(\frac{W^3}{8} + \frac{W^3}{8}\right)\right]$$

$$V_{bi} = \frac{qa}{2\varepsilon_s}\left[\frac{W^2}{4}W - \frac{2W^2}{3(8)}W\right]$$

$$V_{bi} = \frac{qa}{2\varepsilon_s}\left[\frac{W^2}{4}W - \frac{W^2}{4}\left(\frac{1}{3}W\right)\right]$$

For linearly graded junction:



$$\mathcal{E}(x) = -\frac{qa}{\varepsilon_s}\left(\frac{(W/2)^2 - x^2}{2}\right)$$

$$\mathcal{E}_m = \frac{qaW^2}{8\varepsilon_s} \quad \rightarrow [a] = cm^{-4}$$

$$V_{bi} = \mathcal{E}_m W - \frac{\mathcal{E}_m W}{3}$$

$$V_{bi} = \frac{2}{3}\mathcal{E}_m W$$

So, at $V = -V_B$ (Breakdown): → $V_B >> V_{bi}$

$$V_{bi} - V = \frac{2}{3}\mathcal{E}W$$

$$V_{bi} + V_B = \frac{2}{3}\mathcal{E}_c W$$

$$V_B = \frac{2}{3}\mathcal{E}_c W$$

$$V_B = \frac{2}{3}\mathcal{E}_c\left(\frac{8\varepsilon_s \mathcal{E}_c}{qa}\right)^{1/2}$$

$$V_B = \frac{4}{3}\mathcal{E}_c^{3/2}\left(\frac{2\varepsilon_s}{qa}\right)^{1/2}$$

* From:

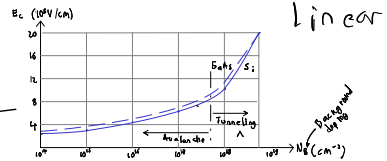
$$V_B \overset{?}{\approx} \frac{\mathcal{E}_c^2 \, \varepsilon_s}{2q N_B}$$

p^+-n

$$V_B = \frac{4}{3} \frac{\mathcal{E}_c^{3/2}}{} \left(\frac{2\varepsilon_s}{q a} \right)^{1/2}$$

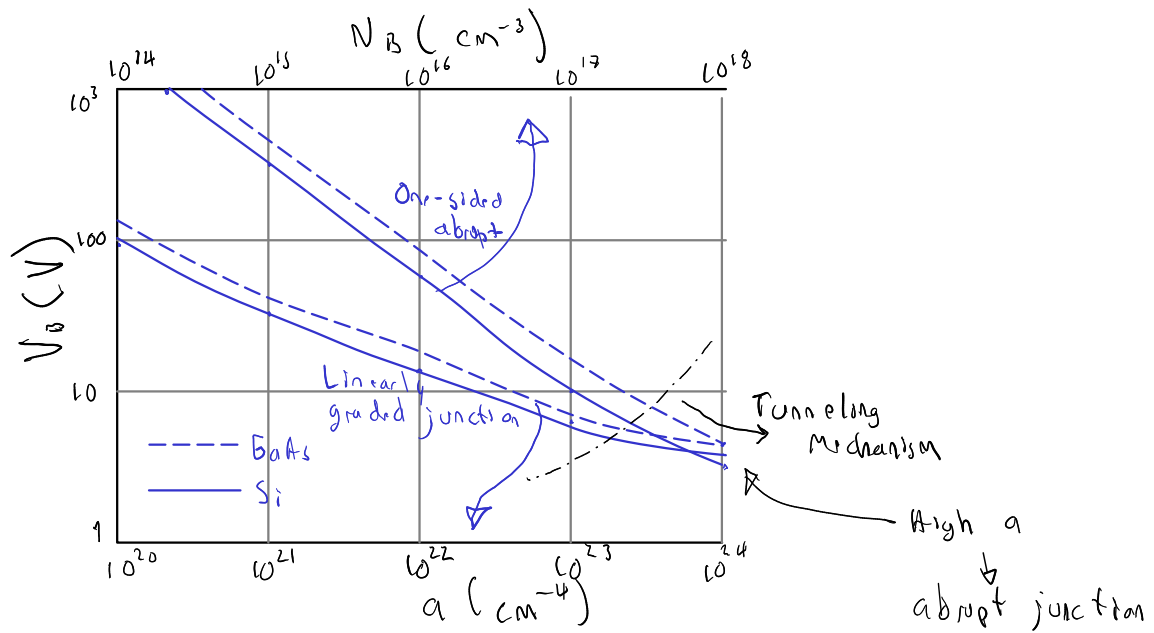
Linearly graded

$+ \quad \mathcal{E}_c(N_B) \leftarrow$



$+ \quad \mathcal{E}_c(a)$ when?

The $V_B(N_B$ or $a)$:



$E_{g_{GaAs}} > E_{g_{Si}}$

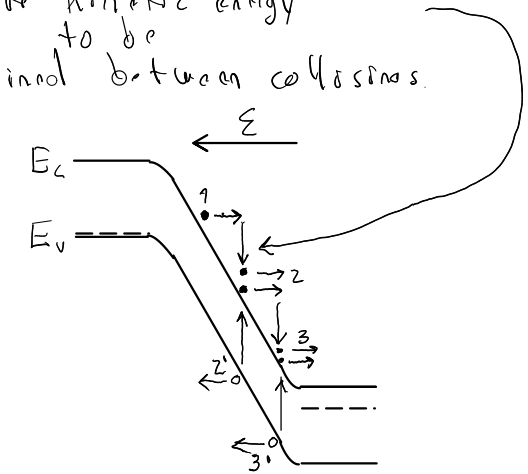
$\downarrow$

$E_{c_{GaAs}} > E_{c_{Si}}$

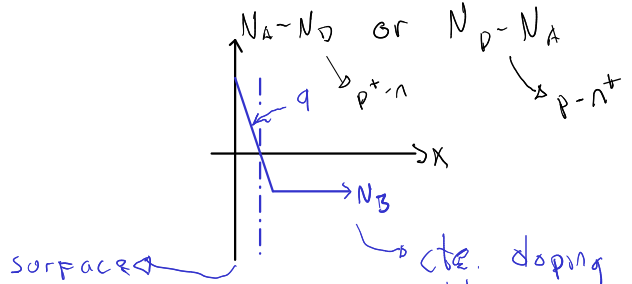
$\downarrow$

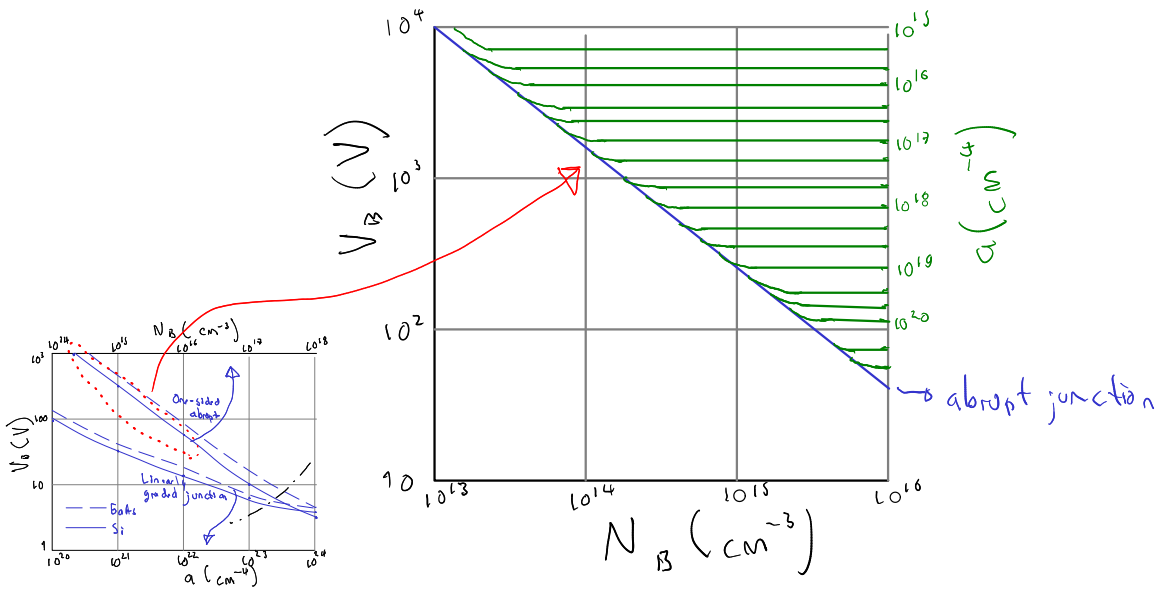
$V_{B_{GaAs}} > V_{B_{Si}}$

GaAs need more kinetic energy to be gained between collisions.



* For diffused junction with a linear gradient near the surface



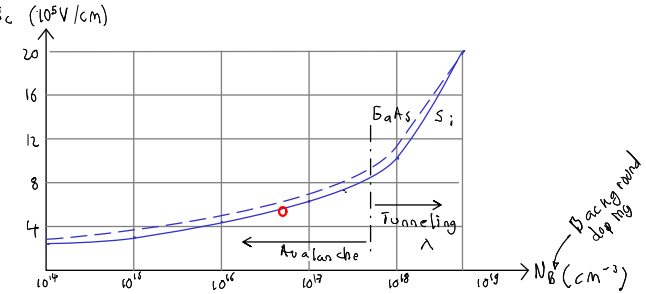


EXAMPLE 8 :

Si : p^+-n , $N_D = 5 \times 10^{16}\ cm^{-3}$ $\quad V_B = ?$

$N_B = N_D = 5 \times 10^{16}\ cm^{-3}$

From :



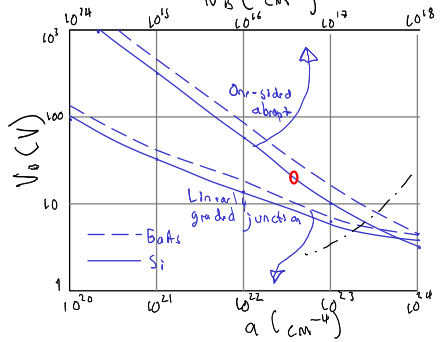
$$E_c\left(5 \times 10^{16}\ cm^{-3}\right) = 5.7 \times 10^5\ V/cm$$

$$\rightarrow V_B = \frac{\mathcal{E}_c W}{2} = \frac{\mathcal{E}_s \mathcal{E}_c^2}{2q N_B}$$

$$V_B = \frac{(11.9 * 5.53 * 10^3\ \cancel{e}/V\cancel{cm})(5.7 * 10^5\ V/\cancel{cm})^2}{2\cancel{e}\,(5 * 10^{16}\ \cancel{cm^{-3}})}$$

$$V_B = 21.4\ V$$

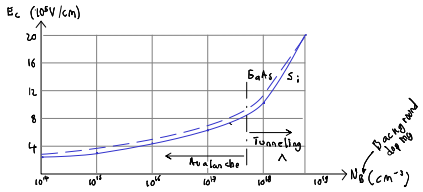
And From:

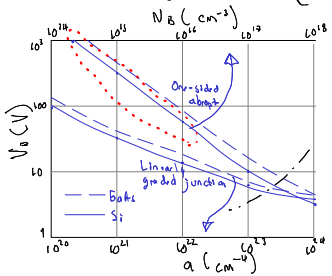


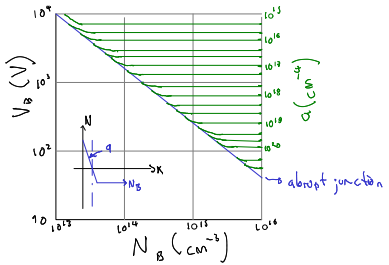
$\Rightarrow V_B \simeq \lg V$

*If at $V_B \to W_m$ and $W_m < W$ ↳ Thick of semiconductor layer

The relation between (N_B or a) and (E_c or V_B):



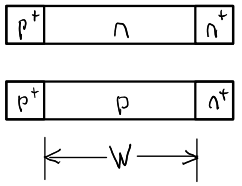




*In the case of $W < W_m$:

↳ Exm: punch-through diode:

depletion layer reach n-n^+ prior to breakdown ←

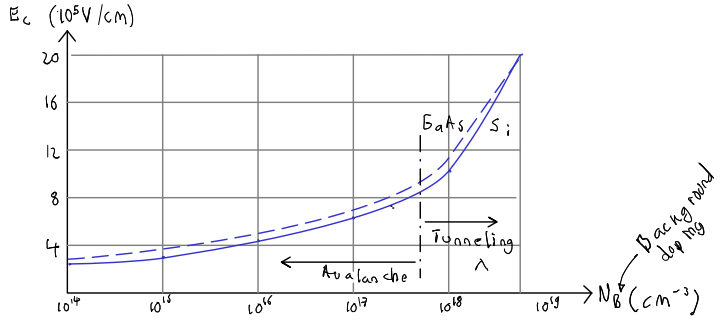


$p^+ - \pi - n^+$

$p^+ - \nu - n^+$

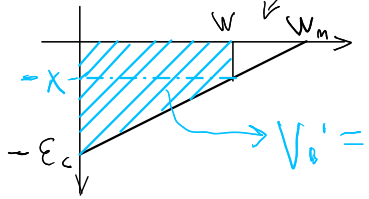
$n\nu$

Same E_c of $W_m < W$:



$W < W_m$

and this E_c:



$$\frac{x}{W_m - W} = \frac{E_c}{W_m}$$

$$x = \frac{E_c}{W_m}(W_m - W)$$

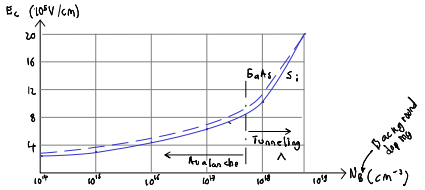
$$V_B' = \frac{W_m E_c}{2} - \frac{(W_m - W)}{2}\frac{E_c(W_m - W)}{W_m}$$

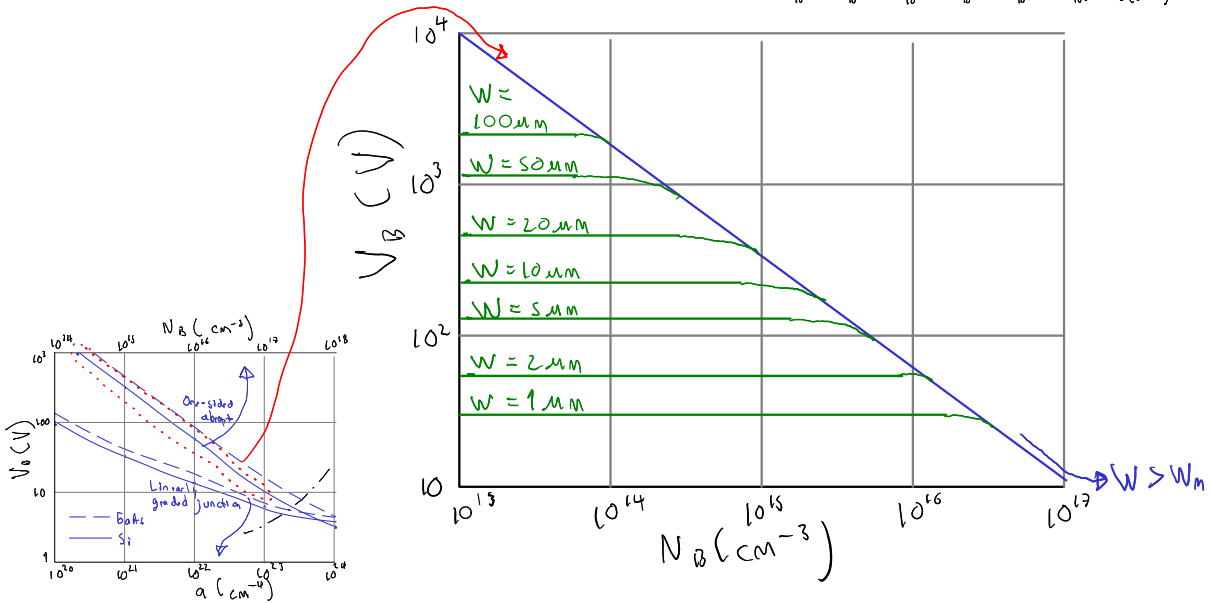
$$V_B' = \frac{E_c W}{2 W_m}(2W_m - W)$$

Thus: $\frac{V_B'}{V_B} = \frac{\frac{\mathcal{E}_c W}{2W_m}(2W_m - W)}{\frac{\mathcal{E}_c W_m}{2}} = \frac{W}{W_m}\left(2 - \frac{W}{W_m}\right)$

$$\boxed{V_B' = V_B \frac{W}{W_m}\left(2 - \frac{W}{W_m}\right)}$$

$$V_B' = \left(\frac{\varepsilon_s \mathcal{E}_c^2}{2qN_B}\right)\frac{W}{W_m}\left(2 - \frac{W}{W_m}\right) \quad \text{and}$$



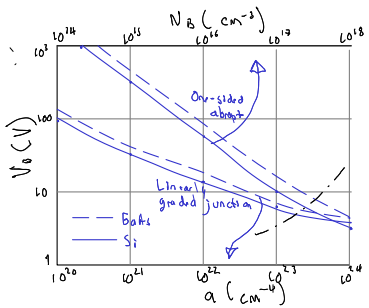


EXAMPLE 9

GaAs p^+-n $N_D = 8 \times 10^{14}\ cm^{-3}$ $W_m = ?$

$W = 20\ \mu m \rightarrow V_B' = ?$

From:



$N_B = N_D = 8 \times 10^{14}\ cm^{-3}$

$\Rightarrow V_B = 500\ V \gg V_{bi}$

So, at: $V = -V_B = -500\ V$

$$W_m = \sqrt{\frac{2\varepsilon_s(V_{bi} - V)}{qN_B}} \simeq \sqrt{\frac{2\varepsilon_s(V_{bi} + V_B)}{qN_B}}$$

$$W_m = \sqrt{\frac{2(12.4 \times 553 \times 10^3\ e/V\,cm)(500\ V)}{(e)(8 \times 10^{14}\ cm^{-3})}}$$

$W_m = 2.93 \times 10^{-3}\ cm = 2.93 \times 10^{1}\ \mu m \simeq 29.3\ \mu m$

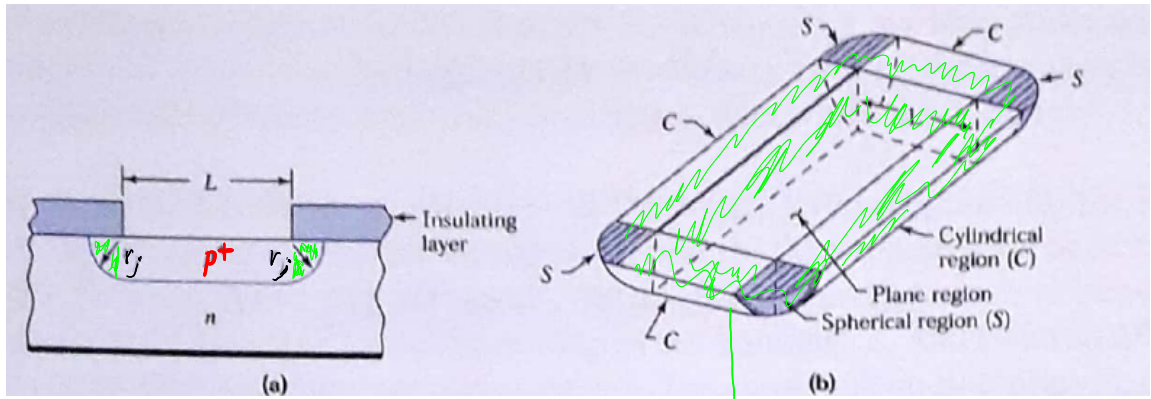
When $W = 20\ \mu m < W_m = 29.3\ \mu m \rightarrow$ punch-through

$\rightarrow V_B' = V_B \frac{W}{W_m}\left(2 - \frac{W}{W_m}\right)$

$$V_B' = (500V)\left(\frac{20\,\mu m}{29.3\,\mu m}\right)\left(2 - \frac{20\,\mu m}{29.3\,\mu m}\right) = 500V\,(0.899)$$

$$V_B' = 449.6\,V$$

* Junction curvature effect:



→ Diffusion through a rectangular mask

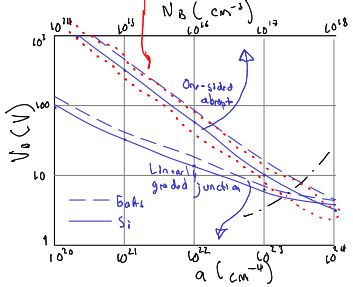
↓

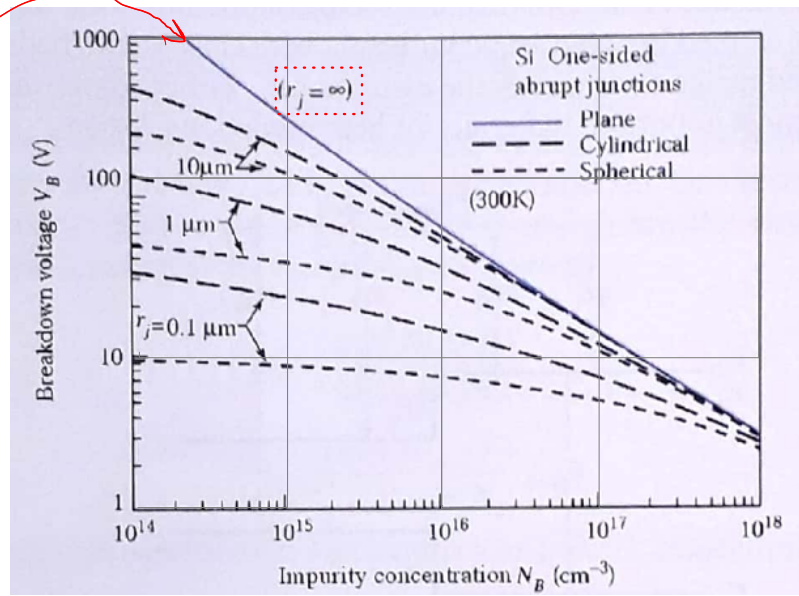
Spherical and cylindrical regions → Higher field

↓

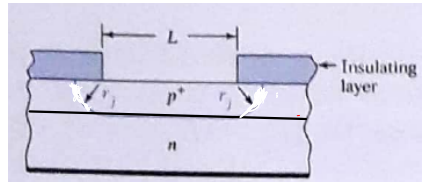
determine the avalanche breakdown voltage

The calculate results for Si p^+-n or p-n^+ at 300K:



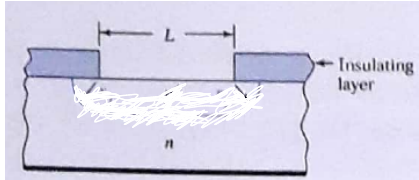


$r_j \to \infty$

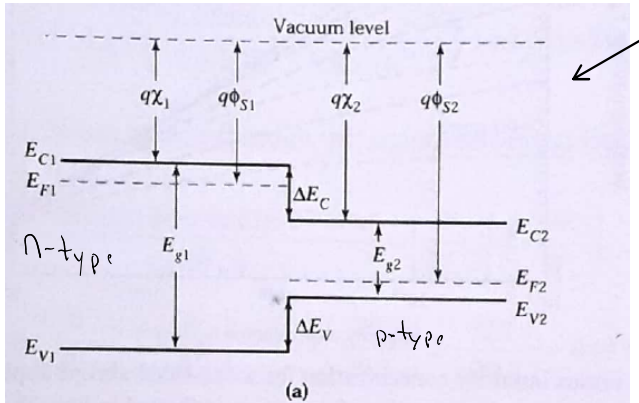


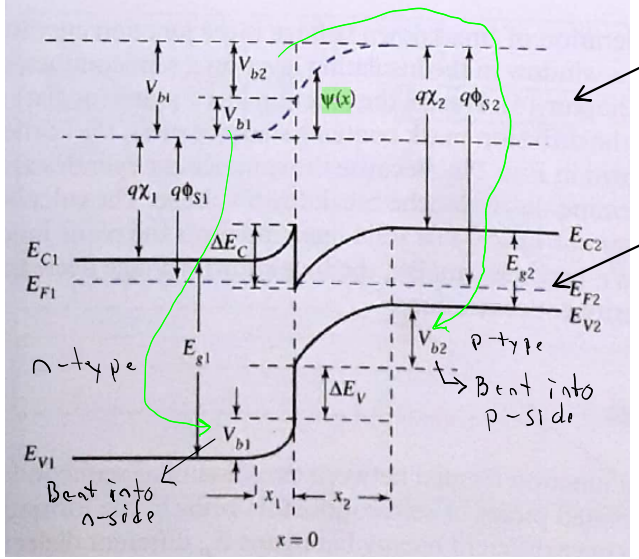
$r_j \to 0 \to$?

$V_B \to 0V$



3.7 HETEROJUNCTION





Two isolated pieces : permitivities.

→ $\varepsilon_1 \neq \varepsilon_2$ → $\Delta E_g = E_{g_1} - E_{g_2}$

→ $\Delta E_c = q(\chi_2 - \chi_1)$

→ $\Delta E_v = E_{g_1} + q\chi_1 - (E_{g_2} + q\chi_2)$

$\Delta E_v = \Delta E_g - \Delta E_c$

Abrupt heterojunction

↳ No graded junction

At thermal equilibrium

Closely matched lattice constants (a) of semiconductors

↓

negligible traps → negligible generation-recombination centers at the interface. Exm: $Al_xGa_{1-x}As$ $(0 \leq x \leq 1)$

When: $x=0$ → GaAs : $E_g = 1.42\,eV$, $a = 5.6533\,\text{Å}$

When: $x=1$ → AlAs : $E_g = 2.17\,eV$, $a = 5.6605\,\text{Å}$

So, for $Al_xGa_{1-x}As$:

χ → E_g

a mismatch is only 0.9% → a cte.

Nondegenerated semiconductors → E_F into E_g → $q\chi$ unaffected by doping

↓

ΔE_c unaffected by doping

ΔE_v unaffected by doping

→ Bent into n-side : V_{b_1} → Bent into p-side : V_{b_2}

→ $V_{bi} = V_{b_1} + V_{b_2}$

→ Free-carrier flux density (free-carrier flow through a area) are continous at the heterointerface

↓

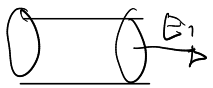
conventional depletion approximation for Poisson's eq.

→ Continuity of electric displacement: $\varepsilon_1 E_1 = \varepsilon_2 E_2$

↑

at $x = 0$ (interface)

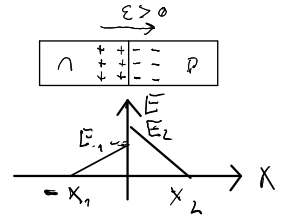
$N_1 x_1 = N_2 x_2$



$\frac{dE}{dx} = \frac{\rho}{\varepsilon}$

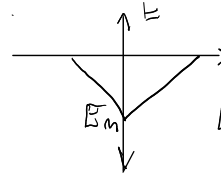
$\frac{d^2V}{dx^2} = -\frac{\rho}{\varepsilon}$

$\frac{d^2V}{dx^2} = -\frac{Nq}{\varepsilon}$



$V_{b_1} = \frac{x_1 E_1}{2}$

For homojunction:



$E_m = -\frac{q N_A^- x_p}{\varepsilon_s}$

$= \frac{q N_D^+ x_n}{\varepsilon_s}$

→ $V_{b_1} = \frac{x_1}{2}\left(\frac{N_1 x_1}{\varepsilon_1} q\right) = \frac{x_1^2 N_1}{2\varepsilon_1} q$

→ $V_{b_2} = \frac{x_2^2 N_2}{2\varepsilon_2} q$

→ $V_{bi} = \left(\frac{x_1^2 N_1}{\varepsilon_1} + \frac{x_2^2 N_2}{\varepsilon_2}\right)\frac{q}{2}$

It is a distinction for V_{b_1}, V_{b_2}, x_2, x_1

$\int_E^{E_1} dE = \int_x^0 \frac{N_1 q}{\varepsilon_1} dx$

$E_1 - E = -\frac{N_1 q}{\varepsilon_1} x$

$E = \frac{N_1 q x}{\varepsilon_1} + E_1$

negative charge

$\int_{E_2}^{E} dE = \int_0^x -\frac{N_2 q}{\varepsilon_2} dx$

$E = -\frac{N_2 q}{\varepsilon_2} x + E_2$

at $x = -x_1 \rightarrow \vec{E} = 0$

so:

$$0 = -\frac{N_1 q x_1}{\varepsilon_1} + E_1$$

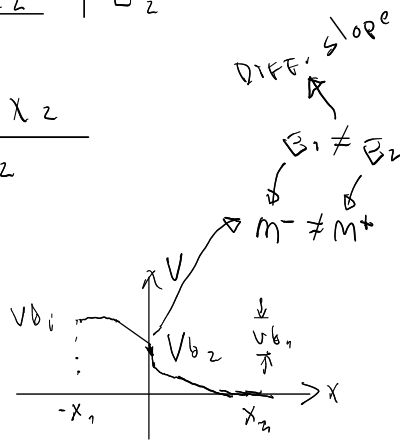
$$E_1 = \frac{N_1 q x_1}{\varepsilon_1}$$

at $x = x_2 \rightarrow \vec{E} = 0$

so:

$$0 = -\frac{N_2 q x_2}{\varepsilon_2} + E_2$$

$$E_2 = \frac{N_2 q x_2}{\varepsilon_2}$$



$$\frac{dV}{dx} = -E$$

$$dV = -E\,dx$$

$$\int_{V_{b_2}}^{V} dV = \int_{x}^{0} \left(-\frac{N_1 q x}{\varepsilon_1} - E_1\right) dx$$

$$V_{b_2} - V = \left(-\frac{N_1 q x^2}{2\varepsilon_1} - E_1 x\right)\Big|_{x}^{0}$$

$$V - V_{b_2} = \left(-\frac{N_1 q x^2}{2\varepsilon_1} - E_1 x\right)\Big|_{0}^{x}$$

$$V = -\frac{N_1 q x^2}{2\varepsilon_1} - E_1 x + V_{b_2}$$

$$\int_{V_{b_2}}^{V} dV = \int_{0}^{x} \left(\frac{N_2 q x}{\varepsilon_2} - E_2\right) dx$$

$$V - V_{b_2} = \frac{N_2 q x^2}{2\varepsilon_2} - E_2 x$$

$$V = \frac{N_2 q x^2}{2\varepsilon_2} - E_2 x + V_{b_2}$$

at $x = -x_1 : V = V_{bi}$

$$V_{bi} = -\frac{N_1 q x_1^2}{2\varepsilon_1} + E_1 x_1 + V_{b_2}$$

$$\rightarrow V_{b_1} = V_{bi} - V_{b_2}$$

$$\rightarrow V_{b_1} = E_1 x_1 - \frac{N_1 q x_1^2}{2\varepsilon_1}$$

at $x = x_2 : V = 0V$

$$0 = \frac{N_2 q x_2^2}{2\varepsilon_2} - E_2 x_2 + V_{b_2}$$

$$V_{b_2} = E_2 x_2 - \frac{N_2 q x_2^2}{2\varepsilon_2}$$

At $x = -x_1$

$$\frac{dV}{dx} = -\frac{N_1 q x}{\varepsilon_1} - E_1$$

$$\frac{dV}{dx} = +\frac{N_1 q x_1}{\varepsilon_1} - E_1$$

$$\frac{dV}{dx} = 0$$

At $x = 0$

$$\frac{dV}{dx} = -E_1$$

At $x = x_2$

$$\frac{dV}{dx} = \frac{N_2 q x_2}{\varepsilon_2} - E_2$$

$$\frac{dV}{dx} = 0$$

At $x = 0$

$$\frac{dV}{dx} = -E_2$$

Discontinuity

$\rightarrow V_{b1} = E_1 x_1 - \frac{N_1 q x_1^2}{2\varepsilon_1}$ $\rightarrow V_{b2} = E_2 x_2 - \frac{N_2 q x_2^2}{2\varepsilon_2}$

$V_{b1} = \left(\frac{N_1 q x_1}{\varepsilon_1}\right) x_1 - \frac{N_1 q x_1^2}{2\varepsilon_1}$

$V_{b1} = \frac{N_1 q x_1^2}{2\varepsilon_1}$ $V_{b2} = \frac{N_2 q x_2^2}{2\varepsilon_2}$

$\rightarrow V_{bi} = \frac{N_1 q x_1^2}{2\varepsilon_1} + \frac{N_2 q x_2^2}{2\varepsilon_2}$

$V_{bi} = \frac{N_1 q}{2\varepsilon_1}\left(\frac{N_2 x_2}{N_1}\right)^2 + \frac{N_2 q x_2^2}{2\varepsilon_2}$

In scr charge conservation

$x_1 N_1^+ = N_2^- x_2$

$x_1 = \frac{N_2 x_2}{N_1}$

$x_2^2 = \frac{V_{bi}}{N_2 q}\left(\frac{2\varepsilon_1\varepsilon_2 N_1}{\varepsilon_1 N_1 + \varepsilon_2 N_2}\right)$

$x_2 = \sqrt{\frac{V_{bi}}{N_2 q}\left(\frac{2\varepsilon_1\varepsilon_2 N_1}{\varepsilon_1 N_1 + \varepsilon_2 N_2}\right)}$ → At thermal equilibrium.

$\rightarrow x_1^2 = \frac{V_{bi}}{N_1 q}\left(\frac{2\varepsilon_1\varepsilon_2 N_2}{\varepsilon_1 N_1 + \varepsilon_2 N_2}\right)$

$x_1 = \sqrt{\frac{V_{bi}}{N_1 q}\left(\frac{2\varepsilon_1\varepsilon_2 N_2}{\varepsilon_1 N_1 + \varepsilon_2 N_2}\right)}$

$\rightarrow V_{b2} = \frac{N_2 q x_2^2}{2\varepsilon_2}$

$V_{b2} = \frac{N_2 q}{2\varepsilon_2}\left[\frac{V_{bi}}{N_2 q}\left(\frac{2\varepsilon_1\varepsilon_2 N_1}{\varepsilon_1 N_1 + \varepsilon_2 N_2}\right)\right]$

$V_{b2} = \frac{\varepsilon_1 N_1 V_{bi}}{\varepsilon_1 N_1 + \varepsilon_2 N_2}$

$\rightarrow V_{b1} = \frac{\varepsilon_2 N_2 V_{bi}}{\varepsilon_1 N_1 + \varepsilon_2 N_2}$ → At T.E.

$\rightarrow V_{bi} = \frac{\varepsilon_1 N_1 V_{bi}}{\varepsilon_1 N_1 + \varepsilon_2 N_2} + \frac{\varepsilon_2 N_2 V_{bi}}{\varepsilon_1 N_1 + \varepsilon_2 N_2}$

$V_{bi} = V_{bi}$ ✓

At non T.E.:

$x_1 = \sqrt{\frac{(V_{bi} - V)}{N_1 q}\left(\frac{2\varepsilon_1\varepsilon_2 N_2}{\varepsilon_1 N_1 + \varepsilon_2 N_2}\right)}$

$x_2 = \sqrt{\frac{(V_{bi} - V)}{N_2 q}\left(\frac{2\varepsilon_1\varepsilon_2 N_1}{\varepsilon_1 N_1 + \varepsilon_2 N_2}\right)}$

$V_{b1} = \frac{\varepsilon_2 N_2 (V_{bi} - V)}{\varepsilon_1 N_1 + \varepsilon_2 N_2}$

$V_{b2} = \frac{\varepsilon_1 N_1 (V_{bi} - V)}{\varepsilon_1 N_1 + \varepsilon_2 N_2}$

▷ EXAMPLE 10

Abrupt heterojunction: $V_{bi} = 1.6V$ $\quad N_1 = \frac{10^{16}}{cm^3} = N_D \quad \varepsilon_1 = 12$

$V_{b_1} = ?$ $\quad x_1 = ?$

$N_2 = \frac{3 * 10^{19}}{cm^3} = N_A \quad \varepsilon_2 = 13$

$V_{b_2} = ?$ $\quad x_2 = ?$

at T.E.

$$\rightarrow V_{b_1} = \frac{(12)(3*10^{19}\,cm^{-3})(1.6V)}{(12)(10^{16}\,cm^{-3}) + (13)(3*10^{19}\,cm^{-3})} = 1.6V \approx V_{bi}$$

lower doping → $V_{b_1} \approx V_{bi}$

$$\rightarrow V_{b_2} = \frac{(12)(10^{16}\,cm^{-3})(1.6V)}{(12)(10^{16}\,cm^{-3}) + (13)(3*10^{19}\,cm^{-3})} = 4.92*10^{-4}\,V \approx 0V$$

$$\rightarrow x_1 = \sqrt{\frac{(1.6V)}{(10^{16}\,cm^{-3})\,e}\left[\frac{2(12)(13)(1.553*10^{3}\,e/V\,cm)(3*10^{19}\,cm^{-3})}{(12)(10^{16}\,cm^{-3}) + (13)(3*10^{19}\,cm^{-3})}\right]}$$

$$x_1 = 46.1*10^{-6}\,cm = 46.1*10^{-2}\,\mu m = 461\,nm$$

lower doping → $x_1 \approx W$

$$\rightarrow x_2 = \sqrt{\frac{(1.6V)}{(3*10^{19}\,cm^{-3})\,e}\left[\frac{2(12)(13)(1.553*10^{3}\,e/V\,cm)(10^{16}\,cm^{-3})}{(12)(10^{16}\,cm^{-3}) + (13)(3*10^{19}\,cm^{-3})}\right]}$$

$$x_2 = 15.4*10^{-9}\,cm = 0.154*10^{-7}\,cm = 0.154\,nm$$

▷ SUMMARY

Depletion region = depletion of mobile carriers